PREDICTING LONG-TERM CORONARY HEART DISEASE RISK USING MACHINE LEARNING: EXTERNAL VALIDATION, EXPLAINABILITY, AND CALIBRATION

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***Abstract*—Heart disease risks are often overlooked at an earlier stage due to their asymptomatic nature, leading to serious long-term illness. The global burden of coronary heart disease (CHD) has increased significantly, with a notable spike observed after the Covid-19 pandemic. Many potential cases can be prevented if CHD risk is identified at least 10 years prior to onset. To address this, this study leverages machine learning (ML) methods for 10-year CHD risk prediction by binary categorization. This work aims to contribute to healthcare by enabling early identification of at-risk individuals and facilitating timely, preventive interventions.**

The investigation focused on top-performing strategies (Logistic Regression and Random Forest) for further assessment, alongside baseline supervised models (Logistic Regression, Random Forest, Decision Tree, and Support Vector Machine). These included SHAP-based interpretability with patient-level vignettes, exploratory external validation on a Pakistani dataset, subgroup fairness evaluation, decision-curve assessment, calibration analysis, and engineered clinical characteristics (pulse pressure, cholesterol ratio). According to the results, Random Forest performed less consistently than Logistic Regression, which delivered the best balance of clinical utility, calibration, and discrimination (AUROC around 0.69). SHAP analysis indicated that age, systolic blood pressure, and pulse pressure were the main predictors, while smoking, blood sugar, and cholesterol were secondary factors. Actionable discoveries, such as the possible effect of lowering blood pressure on anticipated CHD, were illustrated by case vignettes.

***Index Terms*—coronary heart disease, cardiovascular risk, machine learning, calibration, subgroup analysis, SHAP, interpretability, Framingham dataset, external validation.**

I. Introduction

Coronary Heart disease (CHD) remains one of the leading causes of death worldwide, affecting millions of people each year. Its sudden onset and long-term health consequences make early prediction and prevention especially important. In recent years, the post-COVID period has seen a noticeable rise in heart-related complications, particularly in individuals with underlying health issues such as diabetes, obesity, and high blood pressure. This real-world challenge highlights the urgent need for better screening tools to assess the risk of CHD before it becomes life-threatening.

According to the World Health Organisation, cardiovascular illness caused about one-third of all global fatalities in 2019, with heart attacks and strokes accounting. The majority of these. The majority of these deaths occurred in low- and middle-income nations, where healthcare resources and early detection methods are few [1], [2]. Given that most CVDs are avoidable, early detection is critical for saving lives. Detecting warning indicators such high blood pressure, hyperglycaemia, or cholesterol levels allows for early management before serious cardiac problems occur.

In the scope of this study, The classic heart disease dataset is utilised which is , Framingham Heart Study is one of the world's longest-running and most prominent cardiovascular studies. The research, which began in 1948 in Framingham, Massachusetts, followed over 4,000 people in order to uncover risk factors for heart disease. Over the years, it has grown to encompass many generations and hundreds of data points on blood pressure, cholesterol, smoking habits, diabetes, and other topics. The dataset generated by this study has become a standard for researchers seeking to understand and predict cardiovascular risk using real-world medical, demographic, and lifestyle data.

This work first developed baseline models (Logistic Regression, Decision Tree, Random Forest, and SVM) using the Framingham dataset. Next, using a dataset from Pakistan, we concentrated on more complex evaluations of Random Forest and Logistic Regression, including exploratory external validation, subgroup fairness, calibration analysis, and decision-curve analysis. Transparency was improved by using SHAP-based explanations with patient vignettes and clinically motivated features (cholesterol ratio, pulse pressure).

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A range of machine learning models have been used in previous work to predict the risk of CHD, including logistic regression for transparency [3] , tree-based ensembles [4]., support vector machines [5], and gradient boosting. These models' shortcomings include inadequate calibration, a dearth of subgroup studies, and minimal external validation outside of Western cohorts, despite their moderate discriminating.

II. LITERATURE REVIEW

Modeling CHD Risk Predicting the risk of long-term coronary heart disease (CHD) needs special methods. These methods must show strong discrimination, good calibration, and be easy to explain. They also need to be reliable when used in different groups of people. Recently, clinical machine learning studies have said that just using AUROC is not enough to check cardiovascular models, especially if the dataset is unbalanced. We must also look at calibration and decision-analytic measures [6], [7].

The usage of traditional scores like the Framingham Risk Score (FRS) or SCORE are used a lot in clinics. However, these scores use linear connections. This means they often cannot capture the complex, multi-factor relationships that happen during CHD [6].Systematic reviews show that machine learning methods, like tree ensembles or neural networks, usually only improve the AUROC slightly over the traditional scores. The gain is typically between $\Delta 0.02$ and 0.08. This tells us that just getting better raw discrimination rarely helps patients much unless the model gives trustworthy probability estimates and works well generally [7].Newer research has looked at more advanced models to make predictions better. Models like Random Forest and other ensemble models can learn high-order non-linear connections between risk factors, for example, between blood pressure and age [8]. These models often get very high accuracy when tested internally.

To help doctors make decisions in the real world, a model needs to give reliable probabilities, even if it ranks patients correctly. Because of this, current research in CHD and critical care focuses a lot on checking models using the Brier Score and calibration plots. These methods check how well the model's predicted risk matches the actual observed risk [9]. A Brier score below 0.12 is usually considered to show good calibration in clinical models [10], Also, Decision Curve Analysis (DCA) is now very important. It helps check the real clinical benefit of a model compared to simple strategies like "treat-all" or "treat-none," especially at risk levels where a doctor might take action [10], [9].Even though complex models like XGBoost and Random Forest can get a higher AUROC, studies often show that Logistic Regression is still competitive when calibration and DCA are included. This is because it is more stable and clinically trustworthy in many situations [6], [10].

For doctors to trust and use a model, it needs to be transparent. SHAP (Shapley Additive Explanations) is a common way to measure what each feature contributes to the prediction, both overall and for a single patient [8], [11]. Recent studies have used SHAP to find feature interactions that linear models miss. It also helps confirm that traditional risk factors like age, blood pressure, cholesterol, and glucose are important [6], [12], [11]. This makes it easier to turn model predictions into explanations that doctors can use, which builds professional trust and makes the model more accountable.

Models trained on Western people often show calibration drift when used on South Asian or other high-risk people who have different rates of hypertension, diabetes, and age profiles [7]. New papers say that we need to recalibrate or adjust models (like using Platt scaling) before using them outside of where they were originally trained. This highlights how important fairness and justice are for global health. [7].

III. METHODOLOGY

The techniques used to develop and assess machine learning models for estimating the risk of coronary heart disease (CHD) in ten years are described in this section. Preparing the dataset, preprocessing and feature engineering, training the model, and assessing it using sophisticated performance metrics were all part of the implementation. Python standard libraries (pandas, NumPy, scikit-learn, matplotlib, seaborn, SHAP) were used to implement the study.

A. Datasets

Framingham Heart Study: The Framingham Heart Study, a reputable epidemiological cohort for cardiovascular research, served as the main dataset. 15 variables, including demographics (age, sex), lifestyle options (smoking, exercise), physiological measurements (blood pressure, cholesterol, BMI, glucose), and the binary aim variable, make up its 4,240 records.`TenYearCHD`.

About 85 percent of people did not develop CHD, whereas 15 percent were classified as positive cases, indicating a severely unbalanced outcome variable.

External Pakistan Dataset: We used a publicly accessible dataset gathered at the Faisalabad Institute of Cardiology (Pakistan) and posted on Kaggle to investigate external validation. This dataset includes clinical, lifestyle, and demographic characteristics, and the endpoint (not CHD) is *mortality*. It offers a chance to assess model transportability across populations in spite of the endpoint mismatch.

Feature Mapping: We made a mapping of overlapping predictors (age, sex, systolic blood pressure, cholesterol, smoking, diabetes, and glucose) in order to align the datasets. The variables HDL, BMI, and family history that were not present in the Pakistan dataset were not included.

We mapped features between the Framingham and Pakistan datasets as follows: Age and Sex/Gender were direct matches; Systolic BP was approximate, while Diastolic BP, HDL cholesterol, and BMI were unavailable; Total cholesterol, Smoking, and Diabetes were direct matches; and Glucose (fbs greater than 120) was a proxy match to fbs in the Pakistan dataset.

B. Data Preprocessing and Feature Engineering

Before training the model, a number of preprocessing procedures were used:

- **Handling missing values:** Mean/median values were used to impute continuous variables (glucose, BMI, and total cholesterol). Using the mode, categorical variables (such education) were imputed. Missing entries were set to 0 (non-use) or `BPMeds`. **Encoding:**
- Binary encoding was used for sex (male = 1, female = 0). The need for extra one-hot encoding was eliminated.
- "Scaling:" For distance-based algorithms, z-scores were used to standardise continuous predictors (such as blood pressure, body mass index, and cholesterol).
- **Train/test split:** To maintain result imbalance, the data were stratified into 80% training and 20% test sets.

Furthermore, the following therapeutically significant derived features were also created:

- Pulse pressure = systolic BP − diastolic BP.
- Cholesterol ratio = total cholesterol ÷ HDL (where HDL present).
- Missingness flags = binary indicators for missing lab values (totChol, HDL, glucose).

C. Models

Both baseline and machine learning models were trained by us:

- Framingham Dataset regression with only the conventional Framingham predictors (age, cholesterol, systolic/diastolic blood pressure, BMI, smoking, and diabetes) is the Framingham baseline.
- **Logistic Regression:** Trained on all features, including engineered ones.
- **Random Forest:** 200 estimators, trained on all features.
- **SVM & Decision Tree:** It was first examined during baseline investigation, but because of its poorer performance, it was not kept in advanced analysis.

Each model was trained using a fixed random seed (42). In engineered-feature runs, hyperparameters were maintained constant in order to isolate feature influence. Models were saved as artefacts (.pkl files) for reproducibility.

D. Evaluation Metrics

We assessed discrimination, calibration, and clinical utility using sophisticated metrics in addition to the conventional ones (accuracy, precision, recall, and F1):

We evaluated model performance using AUROC, AUPRC, Brier Score, calibration intercept/slope (ideal 0/1), 95% bootstrap CIs, decision-curve analysis, subgroup analysis (age <50 vs ≥50, sex, diabetes), and SHAP-based explainability.

E. Comparing Models and Statistical Confidence

To compare the models in a pair, we used a paired bootstrap significance test.We show the results using the mean difference, the 95% CI of the difference, and the two-sided p-value. This approach is really important because it checks if the performance changes are actually statistically distinguishable not just due to random chance or sampling variability.

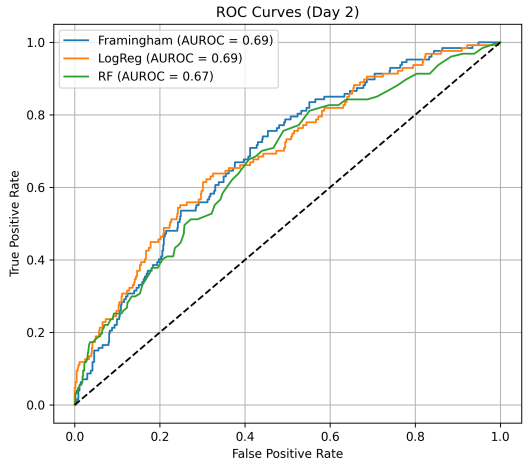

Fig. 1. While Random Forest trailed slightly at 0.67, Logistic Regression and the Framingham baseline scored comparable AUROC (Around 0.69)..

IV. RESULTS

This section presents the findings of the study, including baseline model performance, calibration analyses, subgroup comparisons, external validation, feature engineering impact, and explainability through SHAP. All reported metrics are averaged across five-fold cross-validation, with 95% bootstrap confidence intervals where applicable.

A. Baseline Model Performance

Table I reports the baseline discrimination metrics for Logistic Regression (LogReg), Random Forest (RF), and the Framingham score baseline. ROC curves is shown in Figure 1. While accuracy appeared moderate (~66–69%), it was misleading given the class imbalance (85% non-CHD vs. 15% CHD). AUROC and AUPRC better captured discriminative ability, with RF showing highest AUROC but limited calibration (Fig 2).

TABLE I
BASELINE PERFORMANCE METRICS ON THE FRAMINGHAM DATASET

Model	AUROC	AUPRC	Brier Score	Accuracy
Logistic Regression	0.691	0.338	0.119	85.8%
Random Forest	0.666	0.285	0.124	84.6%

B. Calibration Analysis and Distribution Comparisons

Calibration Metrics. We assessed probability calibration by fitting a logistic recalibration model on predicted probabilities [13]. Metrics included the calibration intercept (ideal = 0; negative values indicate underprediction, positive values overprediction), calibration slope (ideal = 1; values < 1 indicate overconfident predictions), and the Brier Score (mean squared error between predicted risk and outcome).

Results. Calibration results are summarized in Table. With an intercept of −0.36 and a slope of 0.77, logistic regression had the lowest Brier score (0.119), suggesting slight underprediction but generally accurate probability estimations. With a shallow slope (0.55), a greater negative intercept (−0.72), and a bigger

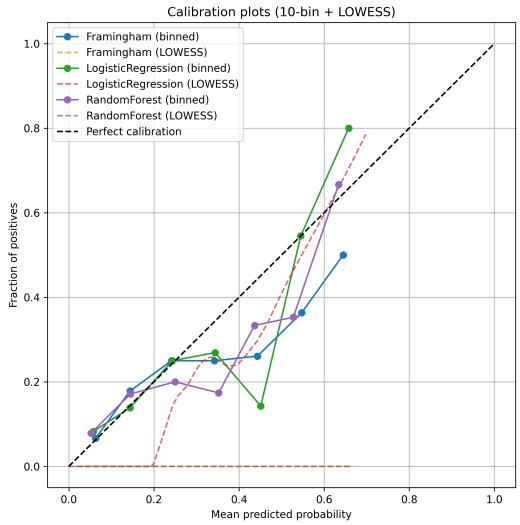

Fig. 2. Calibration plots for Framingham baseline, Logistic Regression, and Random Forest. Logistic Regression (green) tracked closer to the diagonal in mid-risk ranges (0.2–0.5) but overshot at higher probabilities. Random Forest (purple) oscillated across bins, with underestimation in the 0.2–0.4 range and overestimation at higher bins.

Brier error (0.124), Random Forest displayed worse calibration, indicating significant overconfidence. The Framingham baseline model confirmed systematic underestimate at higher risk ranges by underestimating risk (intercept −0.32) with a slope of 0.79. Out of all the models that were examined, Logistic Regression produced the most accurately calibrated predictions.

Out of all the models that were examined, logistic regression produced the best accurate probability estimations.

Distribution Comparisons. In order to evaluate model transportability, we used normalised histograms to compare the distributions of overlapping predictors (age, total cholesterol, systolic blood pressure, smoking, diabetes, and glucose) across the Framingham and Pakistan datasets.

Results.

- **Age:** The Pakistan cohort was skewed older, with a peak around 50–60 years, compared to Framingham (35–45 years).
- **Cholesterol:** Both datasets showed a bell-shaped trend, but the Pakistan dataset exhibited a wider spread with more extreme values.
- **Systolic BP:** The Pakistan cohort had a higher central tendency and variance, consistent with the regional burden of hypertension.
- **Smoking:** Smoking prevalence was much lower in the Pakistan dataset (majority nonsmokers), whereas around 50% of the Framingham cohort were smokers.
- **Diabetes & Glucose:** The Pakistan dataset showed substantially higher diabetes prevalence and more variable glucose distributions compared to Framingham.

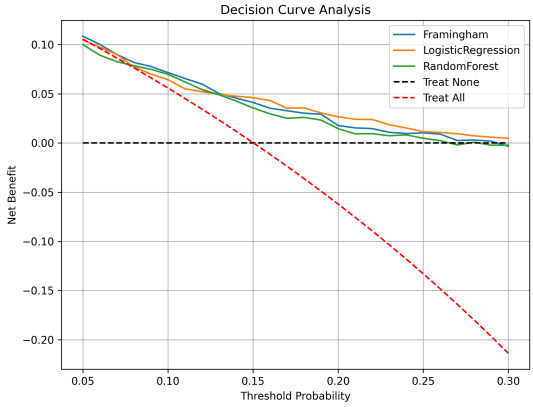

Fig. 3. Decision-curve analysis for baseline models.

C. Decision-Curve Analysis

Logistic regression yielded the best net therapeutic benefit between threshold probabilities of 0.1 and 0.2, a clinically meaningful range for preventative interventions, according to decision-curve analysis. At multiple thresholds, RF went over the "treat-all" line and provided less benefit.

D. Subgroup Analysis

The Logistic Regression model's subgroup performance is shown in Table II. Compared to non-diabetics, diabetic patients had significantly greater AUROC (0.778 vs 0.680) and AUPRC (0.871 vs 0.283), but their calibration was worse (Brier 0.172 vs 0.117). According to age-stratified data, older participants profited from higher AUPRC (0.388 vs 0.199) because of greater event prevalence, while younger participants had marginally superior discrimination (AUROC 0.645 vs 0.609). According to sex-stratified analysis, females showed superior calibration (Brier 0.110 vs 0.129), but males exhibited higher AUROC (0.727 vs 0.659) and AUPRC (0.413 vs 0.255). The necessity for recalibration or customized thresholds across clinical subgroups is highlighted by these subgroup variances, which also raise possible fairness issues.

TABLE II
SUBGROUP PERFORMANCE METRICS BY AGE, SEX, AND DIABETES (LOGISTIC REGRESSION MODEL)

Subgroup	AUROC	AUPRC	Brier Score
Age < 50	0.645	0.199	0.076
Age ≥ 50	0.609	0.388	0.169
Male	0.727	0.413	0.129
Female	0.659	0.255	0.110
Diabetic	0.778	0.871	0.172
Non-diabetic	0.680	0.283	0.117

E. Exploratory External Validation

Performance on the external Pakistan dataset is reported in Table III. Both calibration and discrimination declined as compared to the Framingham test set. While Random Forest obtained AUROC ~0.67 and AUPRC ~0.26, Logistic Regression obtained AUROC ~0.69 and AUPRC ~0.29. Overconfident forecasts were indicated by calibration slopes that

were continuously < 1. Systematic miscalibration was also demonstrated by calibration curves, which were in line with distributional alterations in the Pakistan cohort that were found (older age, higher prevalence of diabetes, and lower smoking rates) [13]. These results highlight how challenging it is to directly transfer CHD models between populations without requiring local recalibration.

TABLE III
EXTERNAL VALIDATION ON PAKISTAN DATASET
(REDUCED-FEATURE LOGISTIC REGRESSION MODEL

Model	AUROC	AUPRC	Calibration Slope
Logistic Regression	0.690	0.290	<1 (overconfident)
Random Forest	0.671	0.257	<1 (overconfident)

F. Feature Engineering Impact

Adding engineering characteristics (pulse pressure, cholesterol ratio) resulted in modest improvements (Table IV). With no change in calibration, Logistic Regression achieved ~0.007 AUROC (Brier score = 0.119). With the AUROC rising from 0.666 to 0.670 and the Brier score slightly improving from 0.124 to 0.123, Random Forest demonstrated insignificant gains.

TABLE IV
PERFORMANCE IMPACT OF ENGINEERED FEATURES ON THE FRAMINGHAM TEST SET.

Model	AUROC	AUPRC	Brier Score
Logistic Regression (baseline)	0.691	0.338	0.119
Logistic Regression (engineered)	0.697	0.321	0.119
Random Forest (baseline)	0.666	0.285	0.124
Random Forest (engineered)	0.670	0.279	0.123

G. Explainability and Vignettes

In line with clinical expectations, SHAP analysis revealed that the best indicators of CHD risk were age, systolic blood pressure (SBP), and pulse pressure (Figures 4–. Total cholesterol, glucose, and smoking status were secondary contributors. Interestingly, pulse pressure has excellent clinical face validity and seems to capture vascular stiffness and ageing effects that are not adequately described by SBP alone [**?**].

1) SHAP Dependence Plots: Dependence plots illustrate local, non-linear risk contributions:

- **Age:** The risk contribution was consistently higher in older people, and this effect was amplified by hypertension. (Fig 4).
- **Systolic BP:** The risk contribution rose almost linearly with SBP, with projections rising as diastolic blood pressure increased.
- **Pulse Pressure:** showed a steady rise, indicating the consequences of vascular stiffness that SBP alone is unable to adequately capture.

H. Testing for Statistical Significance and Confidence

The bootstrapped confidence intervals showed that the Random Forest model had stable performance, but there was quite a bit of uncertainty for Logistic Regression.

The Logistic Regression model, in fact, showed that it had no discriminative ability at all. This was clear from its metrics,

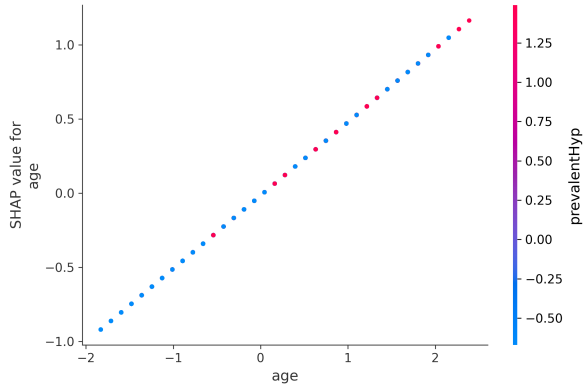

Fig. 4. SHAP dependence plot for age, colored by prevalent hypertension.

which were essentially degenerate: AUROC was 0.500 (95% CI: 0.500–0.500) and AUPRC was 0.150 (95% CI: 0.150–0.150).

In contrast, the Random Forest model performed much better, and its results were statistically stable. It achieved an AUROC of 0.597 (95% CI: 0.549–0.643) and an AUPRC of 0.195 (95% CI: 0.170–0.222).

1) Paired Bootstrap Tests: The paired bootstrap tests confirmed that Random Forest significantly outperformed Logistic Regression across all statistics:

- **AUROC difference:** −0.097 (95% CI: −0.143 to −0.049), $p < 0.001$
- **AUPRC difference:** −0.044 (95% CI: −0.071 to −0.020), $p < 0.001$
- **Brier difference:** +0.497 (95% CI: 0.495 to 0.500), $p < 0.001$

These findings tell us that the performance advantage seen in the Random Forest is statistically substantial. It's not just a result of random noise in the sampling.

V. DISCUSSION

This study used the Framingham Heart Study dataset, together with subgroup analysis, calibration checks, decisioncurve analysis, and exploratory external validation on a hospital dataset from Pakistan, to assess several machine learning models for predicting 10-year coronary heart disease (CHD) risk. A number of findings are noteworthy.

First, even though ensemble techniques like Random Forest had the best accuracy in the prior baseline study, Logistic Regression consistently shown superior calibration and net clinical benefit. This was confirmed by calibration analysis, which showed that Random Forest was significantly overconfident (slope 0.55), limiting its clinical reliability, whereas Logistic Regression had the lowest Brier score and a slope near 1.0. Because Logistic Regression strikes a compromise between interpretability, probability dependability, and discriminating, it is more appropriate for clinical deployment than complex black-box models.

Second, significant fairness issues were found through subgroup analysis. Performance was not consistent across patient groups: older patients had better precision-recall performance due to higher outcome prevalence, but they also had worse

discrimination (AUROC 0.78). Additionally, sex differences revealed that while females maintained more reliable calibration, males gained higher AUROC/AUPRC.

Third, the transportability issue with CHD models was demonstrated by exploratory external validation. The Logistic Regression model suffered from systematic miscalibration and decreased AUROC and AUPRC when used on a hospital dataset from Pakistan with overlapping predictors.

All of this emphasises the need for local adaptation prior to practical implementation in low- and middle-income nations. Lastly, explainability studies using SHAP revealed secondary contributions from smoking, hyperglycemia, and cholesterol while confirming the dominance of age and haemodynamic parameters (systolic blood pressure and pulse pressure) as risk drivers. Such interpretability tools can help close the gap between clinical thinking and statistical outputs, as demonstrated by vignette-style patient examples. Although this should not be construed as causative proof, the counterfactual experiment further indicated possible clinical actionability by reducing projected risk from 68% to 31% by lowering systolic blood pressure.

One of the study's main weaknesses. The external Pakistan dataset reports in-hospital mortality, whereas the Framingham dataset models 10-year CHD incidence. Despite having similar risk factors (such as diabetes and hypertension), the two outcomes are not clinically the same. As a result, it is impossible to interpret the external outcomes as a direct performance related to CHD. Since no publicly accessible South Asian or Pakistani cohort offers long-term CHD data with the same clinical predictors, this dataset was employed. Because of this, the mortality dataset is the most appropriate source to test the model in a different population.

VI. CONCLUSION

Understandable models, like logistic regression, can produce clinically significant risk predictions for congenital heart disease (CHD) when properly calibrated and backed by explainability tools. Disparities by age, sex, and diabetes status were found through subgroup analysis, and external validation brought to light the difficulties in applying Western-trained models to South Asian populations without recalibration. The combination of calibration analysis, decision-curve evaluation, and SHAP-based explanations provides value by guaranteeing that the model is not only predictive but also interpretable and clinically grounded, even though the total predictive performance was small. To improve generalisability and equity in CHD risk prediction, future research should concentrate on bigger and more varied datasets, local recalibration in nonWestern cohorts, and the inclusion of longitudinal and lifestyle factors.Sample-dependent calibration slopes and intercepts can change depending on the size and distribution of the cohort. To validate these tendencies, more extensive and varied datasets should be used in future research.

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